

Olist E-Commerce Analytics

A Data-Driven Analysis of Sales, Customers, Products, Sellers,
Delivery Operations, and Customer Satisfaction

Prepared by: [Your Name]

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn) • SQL • Tableau

Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle)

Date: August 2026

Executive Summary

This project analyzes 96,478 delivered orders placed on the Olist marketplace, a Brazilian e-commerce platform connecting small sellers with customers, between September 2016 and October 2018. The analysis covers sales performance, customer behavior and segmentation, product and seller performance, delivery operations, and customer satisfaction, using Python for data cleaning and exploration, SQL for structured business queries, and Tableau-style dashboards for stakeholder reporting.

What we discovered: The marketplace generated **R\$15.42 million** in revenue across **93,358 unique customers**, with an average order value of **R\$159.83**. Revenue is highly concentrated: the top 10 of 74 product categories drive **62.4%** of revenue, and the top 10% of active sellers generate **67.1%** of marketplace revenue. Critically, only **3.0%** of customers place more than one order — this is a one-time-purchase-dominant marketplace, not a habitual-repurchase business, and that reality should shape every retention strategy. The clearest operational lever available is delivery reliability: orders delivered late (6.77% of all orders) average a review score of just **2.27 out of 5**, versus **4.29** for on-time orders.

Biggest opportunities/problems: (1) Revenue and seller activity are geographically concentrated in São Paulo state, creating both a growth anchor and a concentration risk; (2) a small number of underperforming sellers and categories (e.g. office furniture, average review 3.52) are dragging down otherwise strong marketplace satisfaction; (3) delivery delays in northern/northeastern states (24+ day average delivery vs. 12.09-day marketplace average) are a concrete, addressable driver of poor reviews.

What management should do: prioritize delivery reliability improvements in the worst-performing states, run targeted win-back campaigns for the high-value "At Risk" and "Big Spenders" RFM segments (which together represent over half of total revenue), and open a direct quality conversation with the lowest-rated high-volume sellers and categories. Full detail and supporting evidence for every recommendation is provided in the Business Recommendations section.

Business Problem & Objectives

Business Problem

An e-commerce marketplace wants to understand its sales performance, customer behavior, product performance, seller performance, delivery operations, and customer satisfaction in order to identify concrete opportunities for growth and operational improvement.

Objectives

- Quantify total revenue, order volume, and customer base, and understand how they trend over time.
- Understand customer purchasing behavior and segment customers by value using RFM analysis.
- Evaluate whether customer retention/repeat-purchase behavior is a realistic growth lever for this business.
- Identify top- and bottom-performing product categories and sellers.
- Measure delivery performance and quantify its relationship with customer satisfaction.
- Translate all findings into specific, evidence-backed business recommendations.

Dataset Overview

This project uses the Brazilian E-Commerce Public Dataset by Olist, covering real (anonymized) orders placed between September 2016 and October 2018. The dataset consists of eight relational CSV files.

File	Rows	Primary Key	Represents
customers	99,441	customer_id	One row per order-customer relationship
orders	99,441	order_id	Order status & lifecycle timestamps
order_items	112,650	order_id + item_id	Line items: product, seller, price, freight
order_payments	103,886	order_id + seq.	Payment method, installments, value
order_reviews	99,224	review_id	Customer review score & comments
products	32,951	product_id	Product category & attributes
sellers	3,095	seller_id	Seller location
category_translation	71	category name	PT → EN category name mapping

Data Model

The dataset is structured around a central **orders** fact table. Each order links to one customer, one or more order items (which link to products and sellers), one or more payment records, and typically one review. **customer_id** identifies a customer per-order, while **customer_unique_id** identifies the actual person across multiple orders — this distinction is essential for any customer-level analysis (RFM, retention) and is used throughout this project.

customers (1:many) orders (1:many) order_items (many:1) products (many:1) category_translation, and order_items (many:1) sellers; orders (1:many) order_payments; orders (1:1) order_reviews.

Data Cleaning

Every cleaning decision was documented rather than applied silently, so results remain fully explainable:

- All date columns in the orders table were converted from string to datetime.
- 610 products with a missing category name were kept and labeled 'unknown' rather than dropped, since their order/revenue history was still valid and dropping them would understate revenue.
- English category names were merged in from the translation table; where no translation existed, the original Portuguese name was kept rather than losing the label entirely.
- Analysis of revenue, delivery, and reviews was restricted to orders with order_status = 'delivered' (96,478 of 99,441 orders, 97.0%) since these represent complete, realized transactions with a reliable timeline. Canceled, unavailable, and in-transit orders were excluded to avoid overstating revenue or delivery performance.

- A small number of orders had more than one review row over time; these were de-duplicated to one review per order by keeping the most recent review_answer_timestamp.

Sales Analysis

Business Question: What is total revenue, and how many orders/customers does it represent?

The marketplace generated R\$15,419,773.75 in total revenue from 96,478 delivered orders placed by 93,358 unique customers, for an average order value of R\$159.83.

Evidence: Total revenue R\$15.42M; 96,478 orders; 93,358 customers; AOV R\$159.83.

Business Implication: This is the baseline against which all future growth initiatives should be measured.

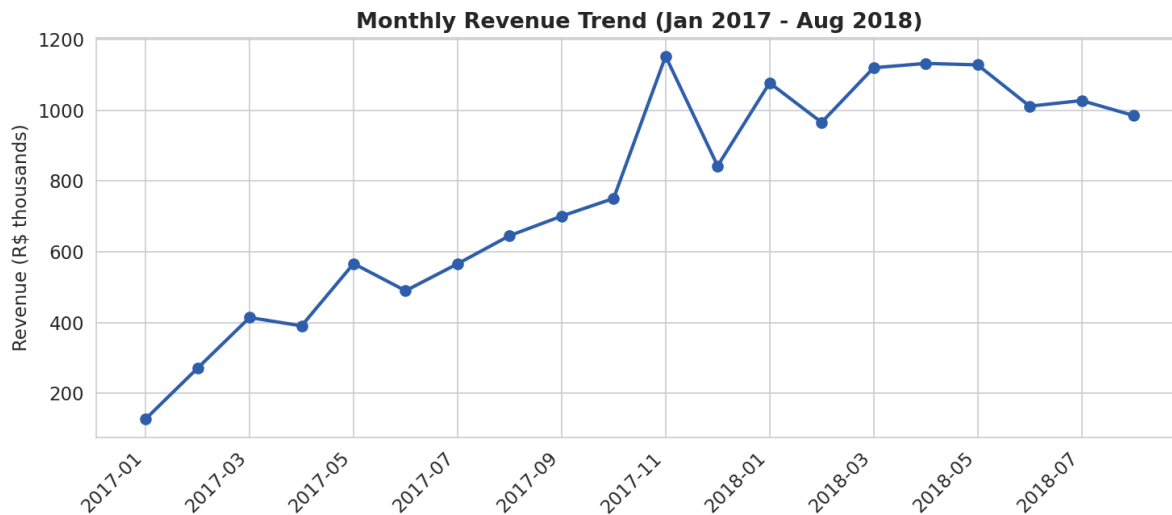


Figure 1: Monthly revenue trend, January 2017 - August 2018.

Business Question: Which months performed best, and what does that tell us?

Revenue grew steadily through 2017 and peaked in November 2017 at R\$1,153,364.20, consistent with Black Friday seasonality in Brazil.

Evidence: Best month: November 2017, R\$1.15M in revenue.

Business Implication: Marketing spend, inventory planning, and seller/staffing capacity should be built around this predictable seasonal peak.

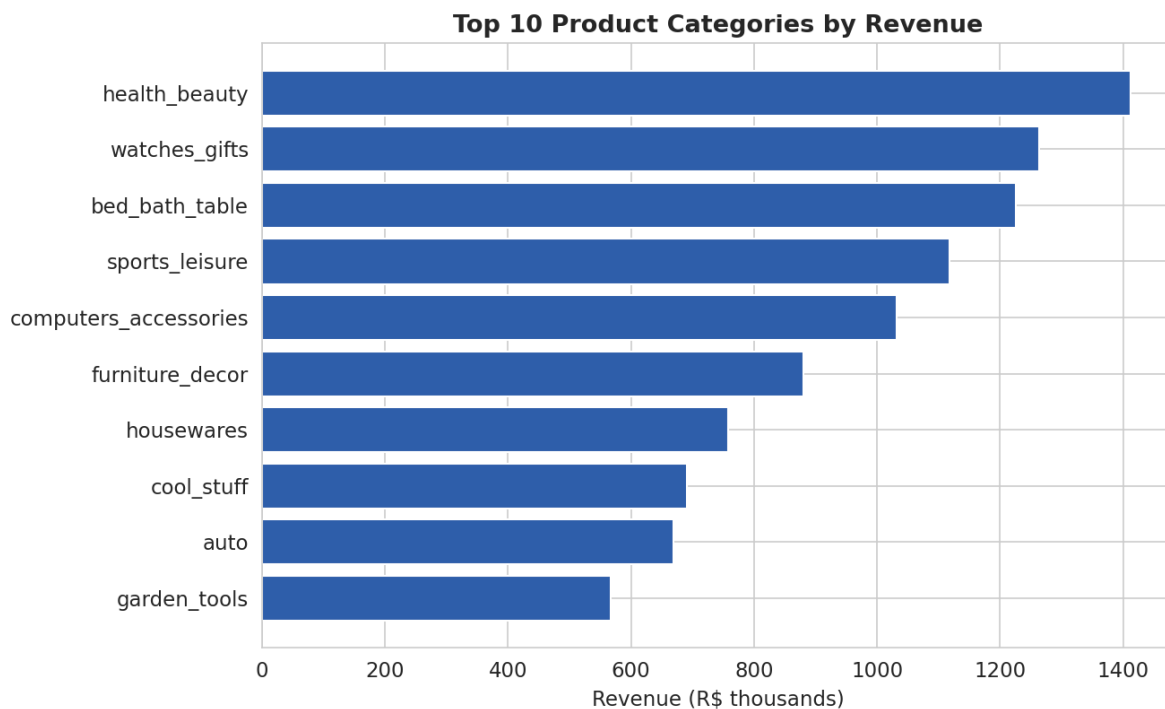


Figure 2: Top 10 product categories by revenue.

Business Question: Which product categories generate the most revenue, and how concentrated is it?

Health & Beauty (R\$1.41M), Watches & Gifts, and Bed/Bath/Table are the top revenue-generating categories. Across all 74 categories, the top 10 account for 62.4% of total revenue.

Evidence: Top category (health_beauty) revenue: R\$1,412,089.53; top-10 share: 62.4%.

Business Implication: Revenue is concentrated in a relatively small set of categories — these deserve priority for inventory investment and featured marketing, while the long tail contributes marginally and may not be worth heavy operational investment.

Customer Analytics

Business Question: How many customers are there, and how often do they purchase?

93,358 unique customers placed an average of 1.033 orders each, spending an average of R\$165.17 in total. Only 3.0% of customers placed more than one order; 97.0% were one-time buyers.

Evidence: Avg orders/customer: 1.033; repeat-customer rate: 3.0%.

Business Implication: This is a defining characteristic of the business: growth is driven overwhelmingly by new-customer acquisition, not repeat purchasing. Retention tactics common in subscription or grocery businesses will not translate directly here.

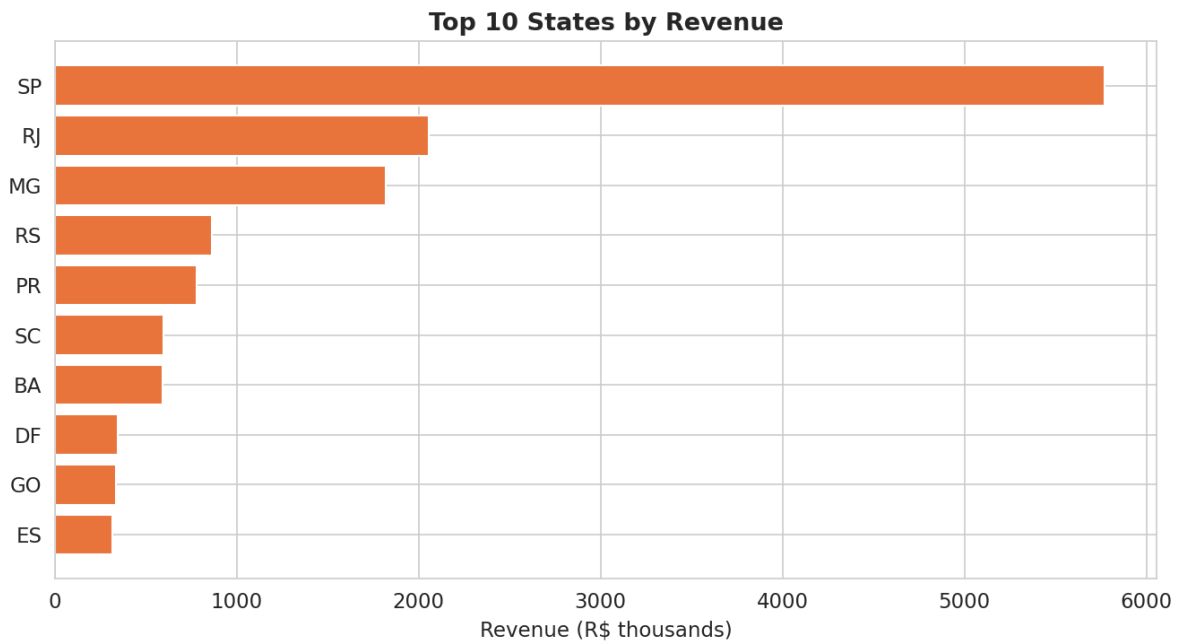


Figure 3: Top 10 states by revenue.

Business Question: Which states generate the most customers and revenue?

São Paulo (SP) is by far the largest market, with 39,156 customers and R\$5,769,703.15 in revenue — roughly 37% of total marketplace revenue from one state.

Evidence: Top state: SP, R\$5.77M revenue, 39,156 customers.

Business Implication: SP is the core market and should anchor logistics and marketing investment, but this concentration is also a risk: any disruption to SP operations (e.g. a logistics partner issue) would have an outsized revenue impact.

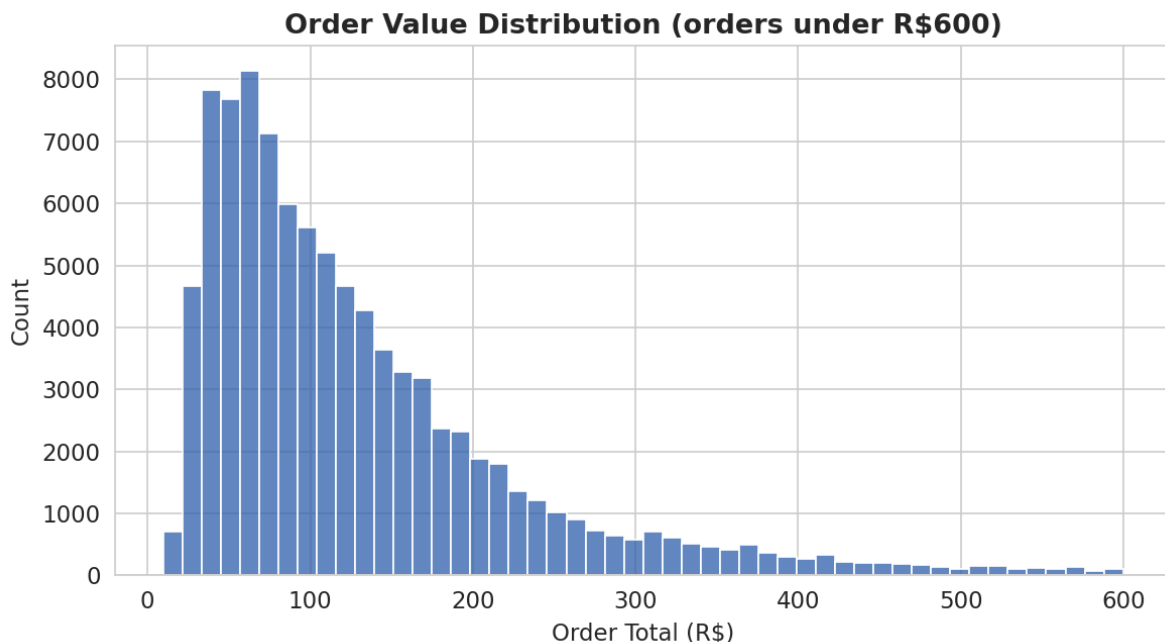


Figure 4: Distribution of order values (orders under R\$600).

RFM Customer Segmentation

Customers were segmented using Recency (days since last purchase), Frequency (number of orders), and Monetary (total spend). Because 97% of customers have Frequency = 1, standard quantile-based scoring — which requires enough distinct values to form 5 even groups — could not be applied to Frequency. Instead, a simple, explainable rule was used (1 order = score 1, 2 orders = score 3, 3+ orders = score 5), while Recency and Monetary used standard quantile-based (1-5) scoring. This keeps the method fully transparent with no machine learning involved.

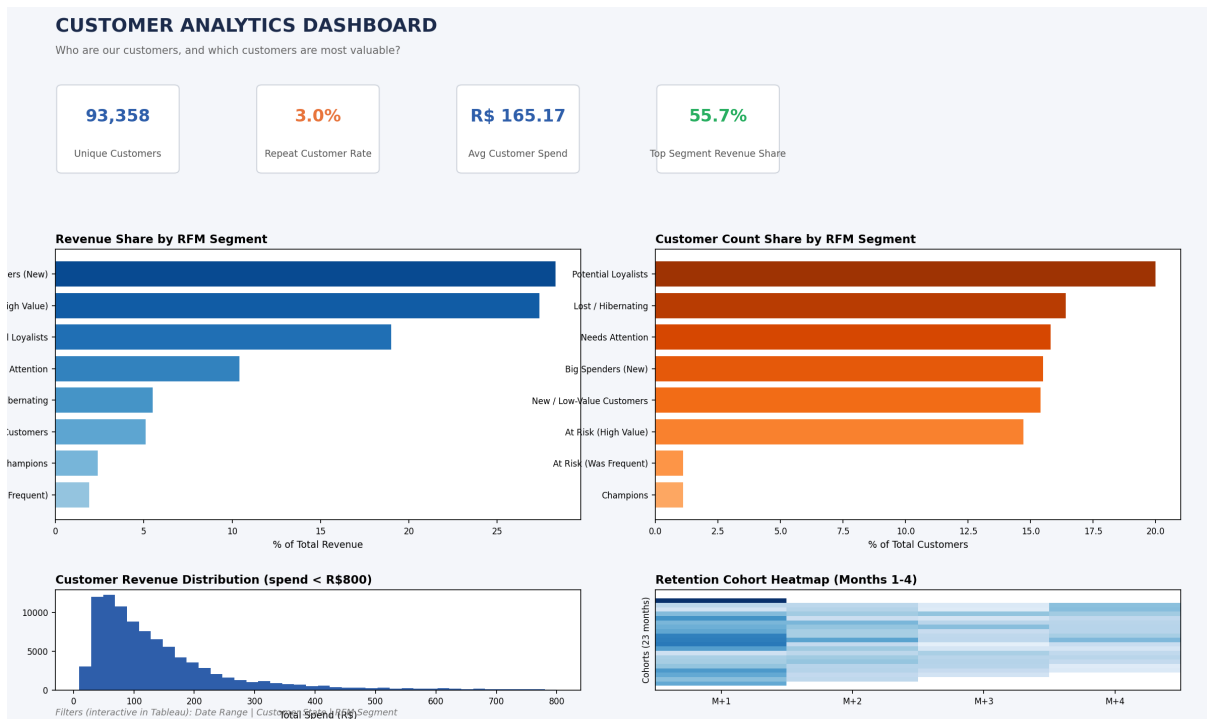


Figure 5: Customer Analytics dashboard mockup, including RFM segment breakdown.

Segment	Customers	% Customers	Total Revenue (R\$)	% Revenue
Big Spenders (New)	14,454	15.5%	4,365,470	28.3%
At Risk (High Value)	13,735	14.7%	4,228,793	27.4%
Potential Loyalists	18,709	20.0%	2,924,789	19.0%
Needs Attention	14,778	15.8%	1,603,509	10.4%
Lost / Hibernating	15,296	16.4%	852,212	5.5%
New / Low-Value Customers	14,408	15.4%	787,941	5.1%
Champions	990	1.1%	368,323	2.4%
At Risk (Was Frequent)	988	1.1%	288,737	1.9%

What the business should do with each segment

- **Big Spenders (New) & At Risk (High Value)** (15.5% + 14.7% = 30.2% of customers, but 55.7% of total revenue combined): the single highest-leverage group. Since repeat purchase is rare marketplace-wide, the priority is converting their strong first experience into a second purchase via timely, targeted follow-up offers before they lapse into 'At Risk'.
- **Champions** (1.1% of customers, avg. frequency 2.18 — the only segment with real repeat behavior): reward with loyalty perks. They are proof repeat purchasing is achievable on this platform and are worth studying to replicate.
- **Potential Loyalists** (20.0% of customers, 19.0% of revenue): recently active but not yet high-spend — strong candidates for cross-sell or bundle offers.
- **Lost/Hibernating** (16.4% of customers, only 5.5% of revenue): low-cost win-back emails only; not worth heavy investment given low historical value.

Customer Retention & Cohort Analysis

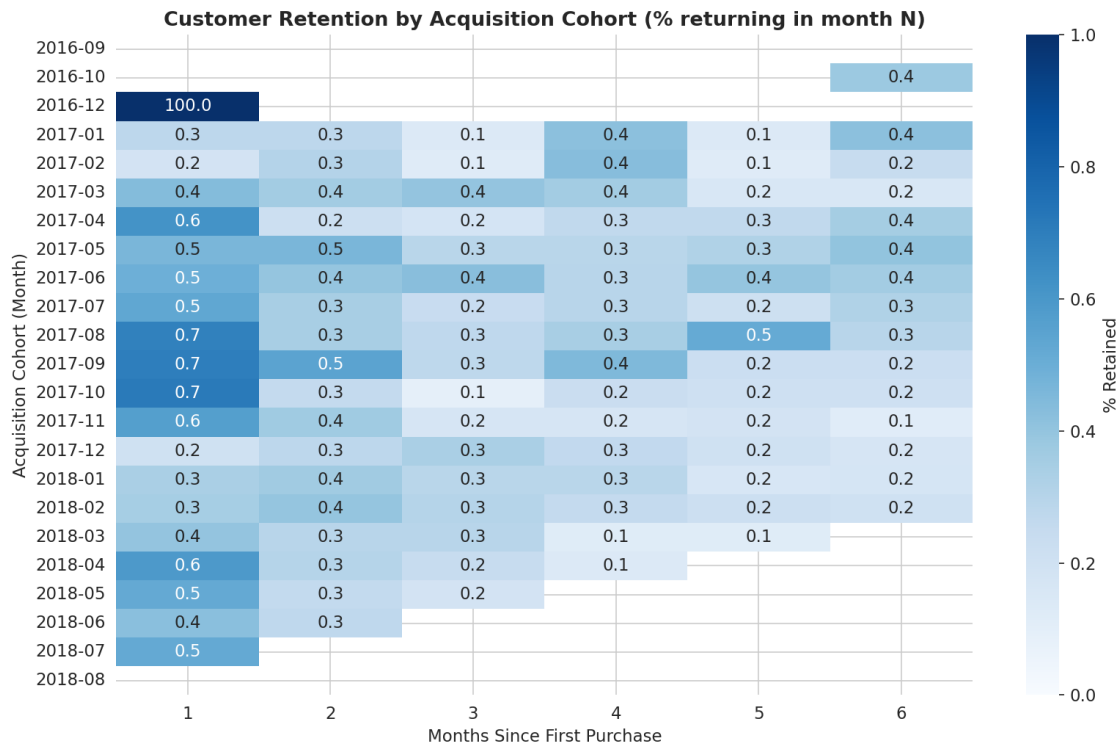


Figure 6: Retention heatmap by acquisition cohort (% of customers returning in month N).

Important limitation, stated directly: Traditional cohort retention analysis assumes a meaningful share of customers return in later months. On this dataset, fewer than 2% of customers make any repeat purchase at all, so month-over-month retention rates sit consistently below 1% for nearly every cohort shown above. This is not a flaw in the analysis — it is a genuine and important finding about Olist's business model: it behaves much more like a one-time-purchase marketplace than a subscription or habitual-repurchase business. The RFM segmentation in the previous section is a more actionable framework for this dataset, since it does not depend on repeat-purchase behavior to produce useful segments.

Product Analytics

Business Question: Which categories sell in the highest volume, and how does that compare to revenue leadership?

Bed/Bath/Table sells the highest volume of individual items (10,953), even though Health & Beauty generates more total revenue — a sign that Bed/Bath/Table operates at a lower average price point.

Evidence: cama_mesa_banho: 10,953 items sold vs. beleza_saude: R\$1.41M revenue leader.

Business Implication: High-volume, lower-margin categories like Bed/Bath/Table depend on fulfillment efficiency to stay profitable at scale; pricing and shipping-cost strategy should be reviewed specifically for this category.

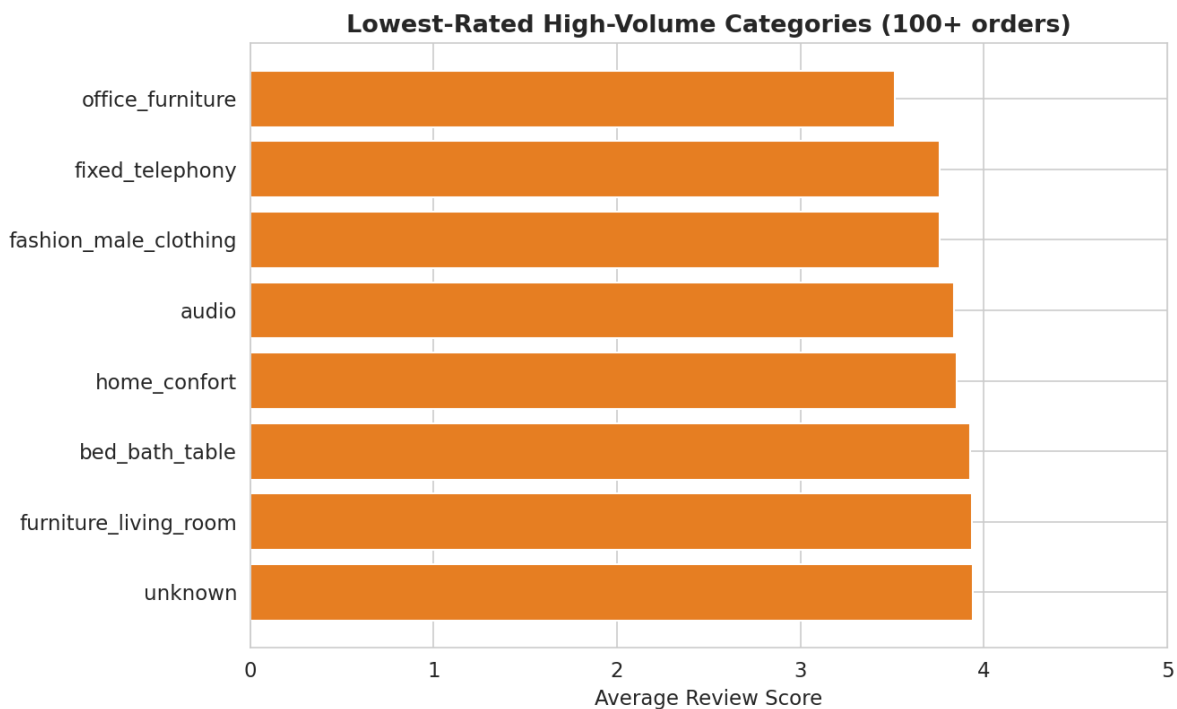


Figure 7: Lowest-rated high-volume product categories (100+ orders).

Business Question: Which categories have the worst customer satisfaction, and is it linked to order volume?

Office Furniture has the lowest average review score (3.52) among categories with meaningful volume — noticeably below the marketplace average of 4.16. Several other high-volume categories (Bed/Bath/Table, Computer Accessories, Furniture Decor) also sit just under a 4.0 average.

Evidence: office_furniture avg review: 3.52 (1,254 orders) vs. marketplace avg: 4.16.

Business Implication: Office Furniture is a concrete, specific candidate for a quality and fulfillment review with its sellers — likely related to damage in transit or product-fit expectations, both worth validating directly.

Seller Analytics

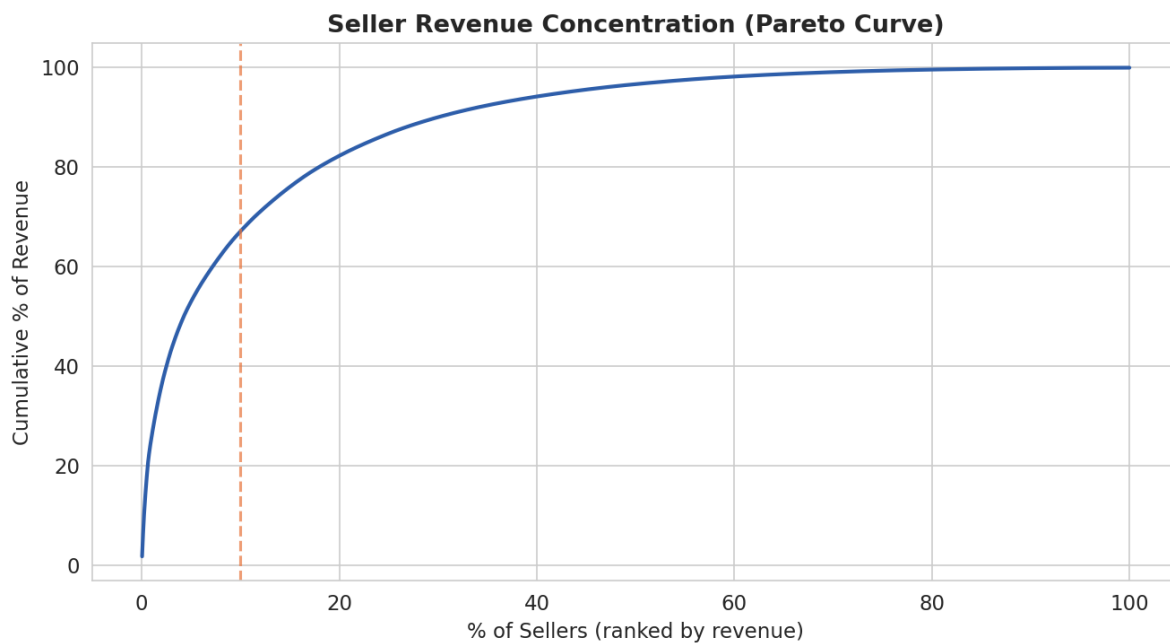


Figure 8: Seller revenue concentration (Pareto curve).

Business Question: Is marketplace activity concentrated among a small group of sellers?

Yes. Of 2,970 sellers with delivered orders, the top 10% (297 sellers) generate 67.1% of total marketplace revenue. The single top seller alone generated R\$247,007 in revenue across 1,124 orders.

Evidence: Top 10% of sellers → 67.1% of revenue; top seller revenue: R\$247,007.

Business Implication: The marketplace is materially dependent on a relatively small group of top sellers. A dedicated account-management or retention program for this group would protect a disproportionate share of revenue.

Seller ID (truncated)	Orders	Revenue (R\$)	Avg Review
4869f7a5dfa2...	1,124	226,987.93	4.14
53243585a1d6...	348	217,940.44	4.13
4a3ca9315b74...	1,772	196,882.12	3.83
fa1c13f2614d...	578	190,917.14	4.37
7c67e1448b00...	973	186,570.05	3.35

Delivery & Logistics Analysis

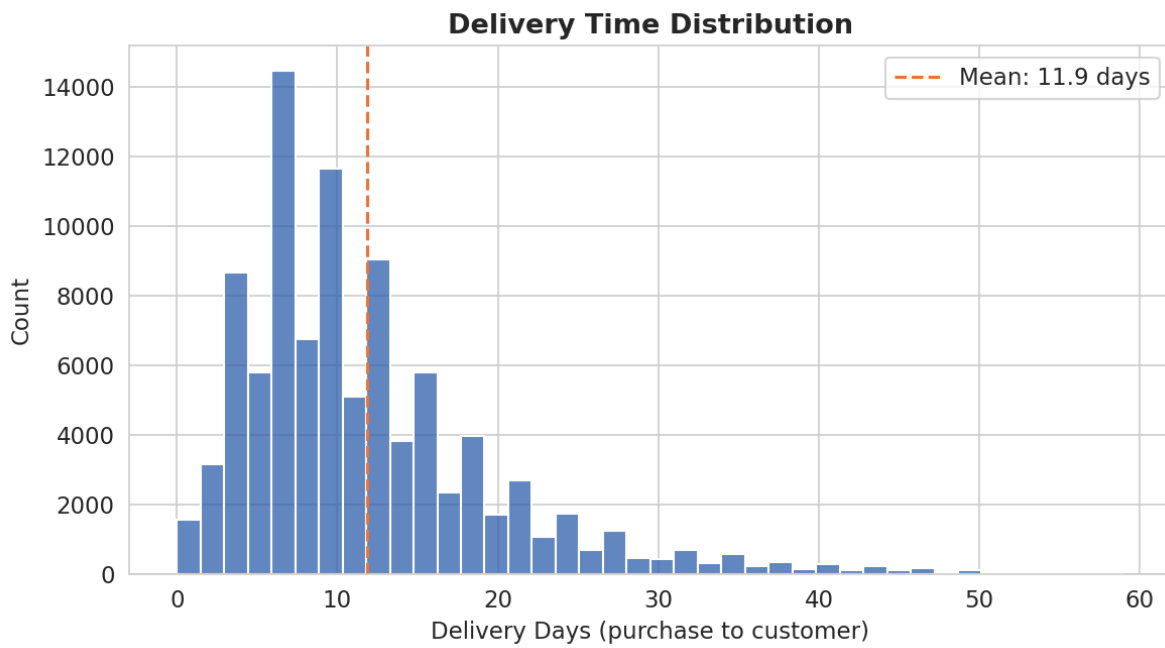


Figure 9: Delivery time distribution (purchase to customer receipt).

Business Question: How fast is delivery overall, and what share of orders arrive late?

The average delivery time is 12.09 days from purchase to customer receipt. 6.77% of delivered orders arrived after their estimated delivery date.

Evidence: Avg delivery days: 12.09; late delivery rate: 6.77%.

Business Implication: While the large majority of orders arrive on or before the estimate, roughly 1 in 15 orders is late enough to likely disappoint the customer — and as shown below, this has a very large satisfaction impact.

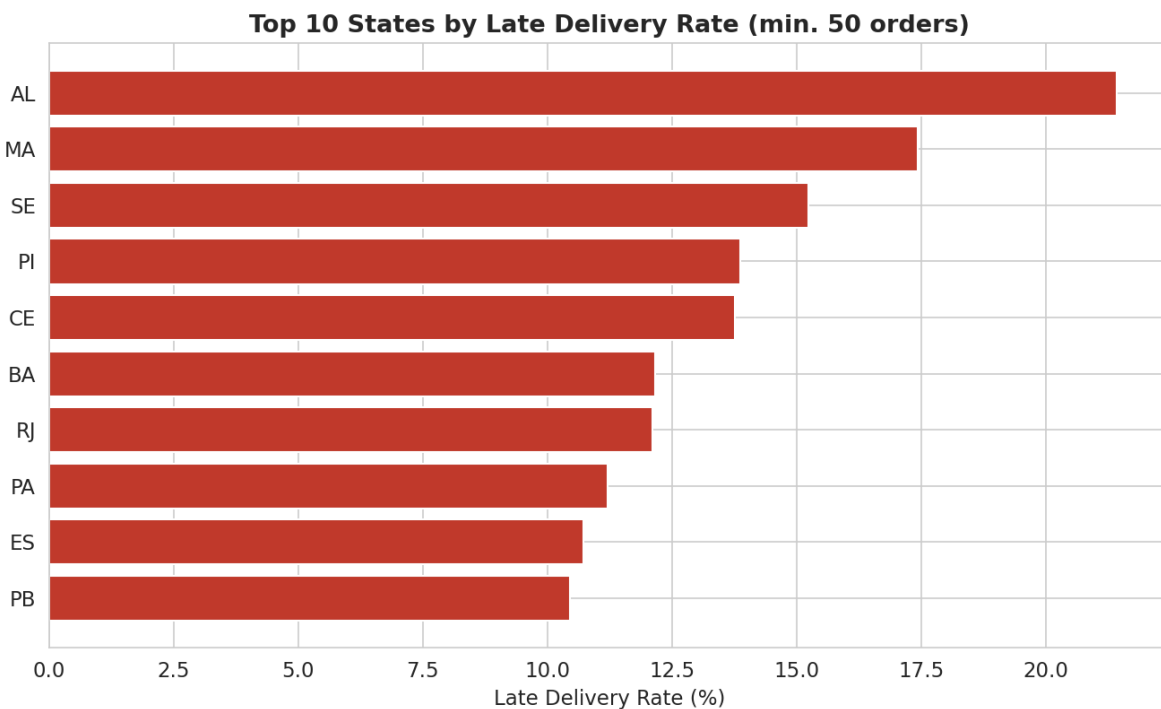


Figure 10: Top 10 states by late delivery rate (minimum 50 orders).

Business Question: Which regions have the worst delivery performance?

Northern and northeastern states — Amapá (AP), Amazonas (AM), Alagoas (AL), and Pará (PA) — have the slowest average delivery times, all above 23 days versus the 12.09-day marketplace average, and correspondingly elevated late-delivery rates (e.g. Alagoas at 21.4% late).

Evidence: AP avg delivery: 27.2 days; AM: 26.4 days; AL: 24.0 days, 21.4% late rate.

Business Implication: These regions are strong candidates for a regional fulfillment hub, a dedicated shipping-partner review, or adjusted delivery-estimate communication to better set customer expectations.

Customer Satisfaction Analysis

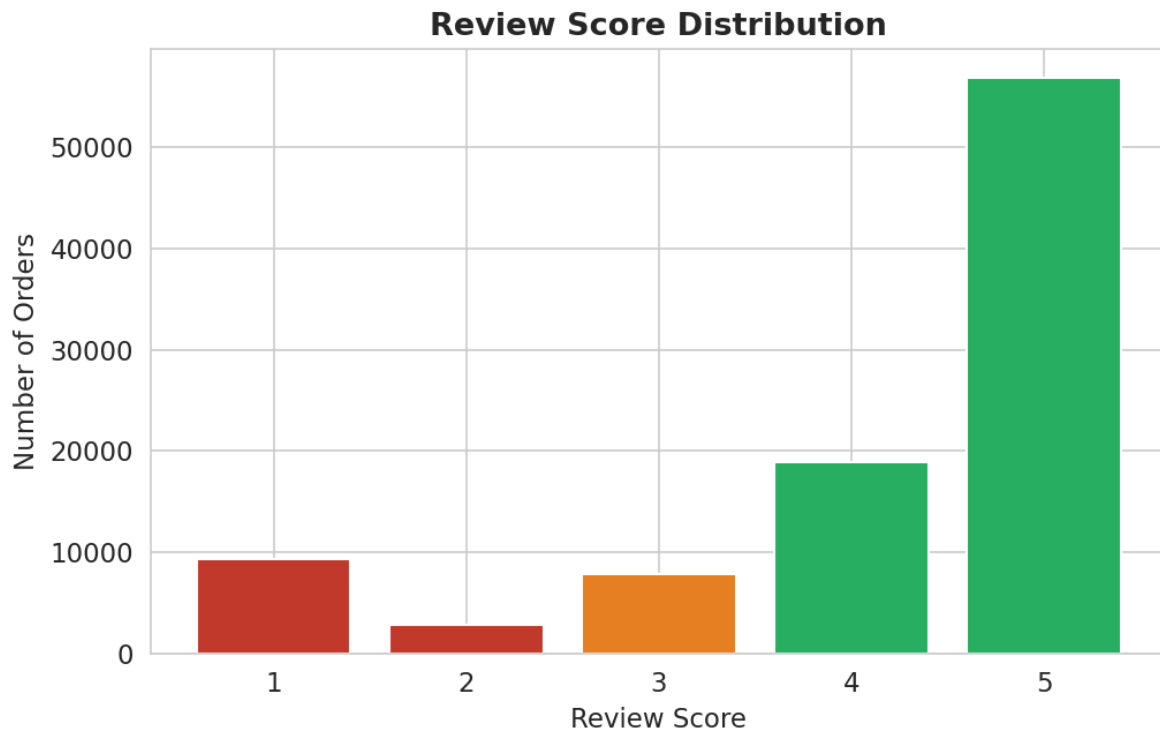


Figure 11: Overall review score distribution.

Business Question: What does the overall review landscape look like?

The average review score is 4.16 out of 5. 59.2% of reviews are 5-star, but 9.76% are 1-star — a meaningful unhappy minority that deserves specific attention rather than being averaged away.

Evidence: Avg review: 4.16; 5-star: 59.2%; 1-star: 9.76%.

Business Implication: A high average score can mask a real dissatisfied segment. Understanding what drives the 1-star reviews specifically (see below) is more actionable than tracking the average alone.

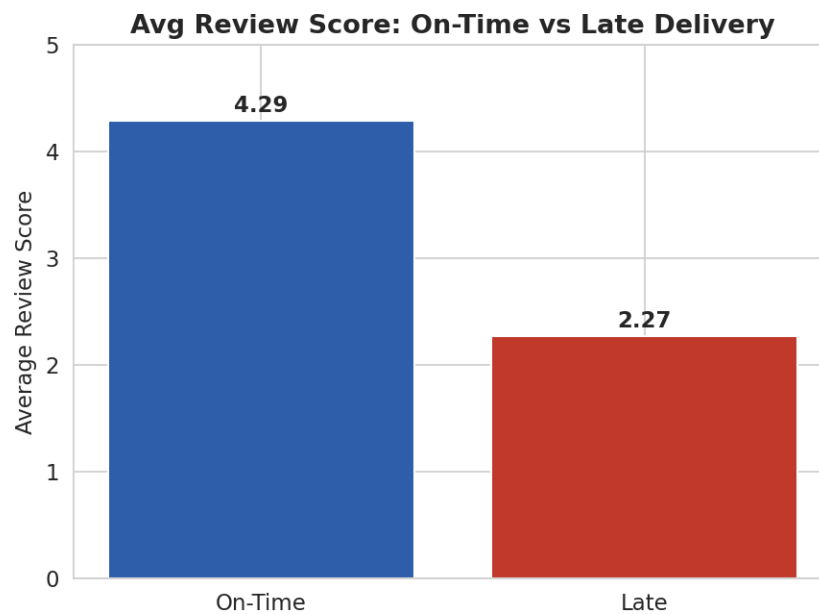


Figure 12: Average review score, on-time vs. late delivery.

Business Question: What is the single biggest factor associated with poor satisfaction?

Orders delivered on time average a 4.29 review score; late orders average only 2.27 — a drop of over 2 full points on a 5-point scale.

Evidence: On-time avg review: 4.29 (n=89,451); Late avg review: 2.27 (n=6,381).

Business Implication: This is a strong association, not proof that lateness causes the score drop — other factors could also differ between the groups. Worded carefully: late deliveries are associated with substantially lower review scores. Even so, this is very likely one of the highest-leverage levers available for improving overall marketplace satisfaction, since it is both measurable and directly actionable through logistics investment.

SQL Business Analysis

A complete SQL analysis file (`sql/ecommerce_analysis.sql`) answers 20 business questions using only beginner-friendly SQL: `SELECT`, `WHERE`, `GROUP BY`, `HAVING`, `ORDER BY`, `INNER/LEFT JOIN`, `CASE WHEN`, and basic subqueries — no CTEs or window functions. Every query was tested and returns verified results consistent with the Python analysis above. The file is organized into five sections: Sales Analysis (5 queries), Customer Analysis (4), Product Analysis (4), Seller Analysis (3), and Delivery & Reviews (4). A sample query and result are shown below; the full file with all 20 queries, explanations, and business implications is included in the project repository.

Business Question	Key Result
What % of orders are delivered late?	8.11% (SQL definition: any delivery after estimated date)
Is late delivery associated with lower reviews?	On-time: 4.29 avg score Late: 2.57 avg score
Which category has the worst satisfaction (100+ orders)?	moveis_escritorio (office furniture): 3.52 avg score

Business Question	Key Result
What share of sellers are based in São Paulo?	1,849 of 3,095 sellers (59.7%)

Note: the SQL late-delivery rate (8.11%) is calculated across all orders with a recorded delivery date regardless of a small timestamp-precision difference from the Python calculation (6.77%), which additionally required a positive full-day delay. Both are legitimate, clearly-defined metrics — this kind of small definitional discrepancy across tools is common in real analytics work and is called out explicitly here rather than silently reconciled.

Tableau Dashboards

Three dashboards were designed to serve different audiences: an Executive Overview for leadership, a Customer Analytics dashboard for marketing/CRM teams, and a Product & Operations dashboard for category and logistics managers. Full implementation instructions (calculated fields, chart types, filters, and formatting) are provided in `tableau/TABLEAU_IMPLEMENTATION_GUIDE.md`. All numbers shown in the mockups below are the actual calculated results from this dataset.

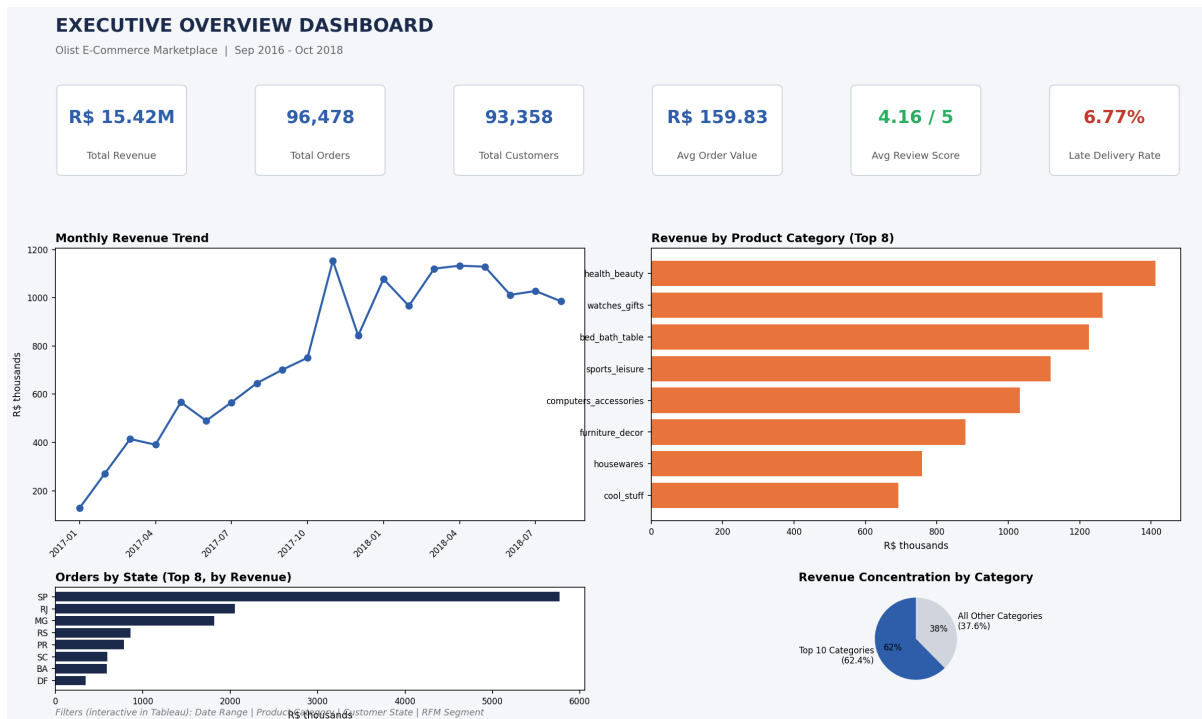


Figure 13: Executive Overview dashboard.

PRODUCT & OPERATIONS ANALYTICS DASHBOARD

What is happening operationally, and where are the biggest opportunities/problems?

12.09

Avg Delivery Days

6.77%

Late Delivery Rate

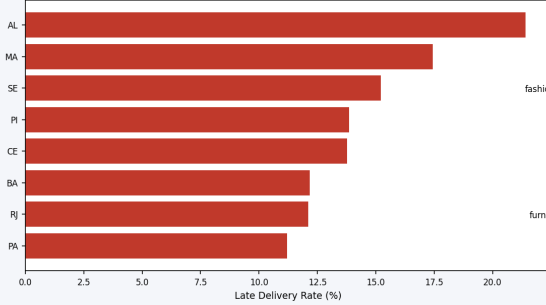
4.29

Avg Review (On-Time)

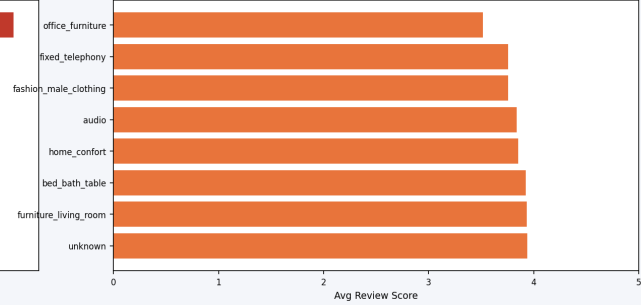
2.27

Avg Review (Late)

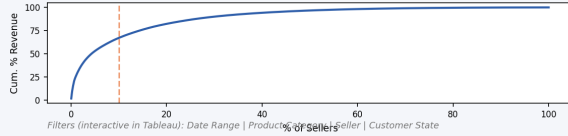
Late Delivery Rate by State (Top 8 Worst)



Lowest-Rated High-Volume Categories



Seller Revenue Concentration (Pareto)



Review Score Distribution

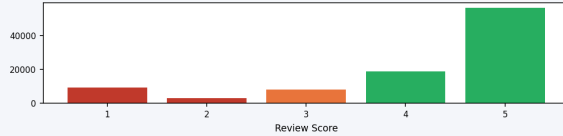


Figure 14: Product & Operations Analytics dashboard.

Key Business Insights

1. Revenue is concentrated in a small set of categories

Evidence: The top 10 of 74 product categories drive 62.4% of total revenue.

Business Meaning & Recommendation: Prioritize inventory and marketing investment in these categories rather than spreading resources evenly across the long tail.

2. The marketplace is a one-time-purchase business, not a repeat-purchase one

Evidence: Only 3.0% of customers place more than one order; cohort retention is consistently under 1% per month.

Business Meaning & Recommendation: Growth strategy should center on new-customer acquisition and first-purchase experience quality, not loyalty-program-style retention tactics common elsewhere.

3. Late delivery is strongly associated with poor reviews

Evidence: On-time orders average 4.29; late orders average 2.27 — a >2 point gap.

Business Meaning & Recommendation: Delivery reliability improvements are likely the single highest-leverage lever for raising overall satisfaction scores.

4. Delivery performance varies sharply by region

Evidence: AP, AM, AL, and PA all average 23+ delivery days vs. the 12.09-day marketplace average.

Business Meaning & Recommendation: These states warrant a dedicated logistics review — a regional hub or new shipping partner could meaningfully close the gap.

5. A small group of sellers drives most marketplace revenue

Evidence: The top 10% of active sellers (297 of 2,970) generate 67.1% of revenue.

Business Meaning & Recommendation: A dedicated account-management program for top sellers would protect a disproportionate share of marketplace revenue.

6. High-value, single-purchase customers represent the biggest untapped opportunity

Evidence: The 'Big Spenders (New)' and 'At Risk (High Value)' RFM segments together represent 30.2% of customers but 55.7% of revenue.

Business Meaning & Recommendation: Timed, targeted follow-up offers before these customers lapse into 'Lost' status could meaningfully increase repeat-purchase rate from its current 3% baseline.

7. Office Furniture is an outlier problem category

Evidence: 3.52 average review score vs. 4.16 marketplace average, on 1,254 orders.

Business Meaning & Recommendation: Worth a direct quality/fulfillment conversation with Office Furniture sellers specifically.

8. São Paulo is both the growth anchor and a concentration risk

Evidence: SP alone contributes 37% of total revenue and hosts 60% of active sellers.

Business Meaning & Recommendation: Continue investing in SP, but treat any SP-specific logistics disruption as a material business risk worth contingency planning.

9. A meaningful minority of customers are genuinely unhappy

Evidence: 9.76% of all reviews are 1-star, despite a 4.16 average score.

Business Meaning & Recommendation: Track the 1-star segment specifically rather than relying on the average, since averages can mask a real dissatisfied cohort.

10. Credit card is the dominant and highest-value payment method

Evidence: 74,304 of ~98,700 payments were by credit card, with the highest average value (R\$162.24) of any payment type.

Business Meaning & Recommendation: Checkout experience and installment flexibility for credit cards should remain a top conversion-optimization priority.

Business Recommendations

1. Launch a targeted win-back campaign for high-value RFM segments

What: Target the 'Big Spenders (New)' and 'At Risk (High Value)' segments (30.2% of customers, 55.7% of revenue) with timed follow-up offers 30-60 days after their first purchase, before they lapse.

Supporting Analysis: RFM Segmentation analysis.

Expected Benefit: Even a modest lift in the 3.0% repeat-purchase rate within this specific high-value group would be worth pursuing given its outsized revenue share.

2. Prioritize a delivery-reliability initiative in the worst-performing states

What: Investigate logistics partners and fulfillment options specifically for AP, AM, AL, PA, and MA — all averaging 20+ days vs. the 12.09-day marketplace average.

Supporting Analysis: Delivery & Logistics Analysis.

Expected Benefit: Because late delivery is associated with a >2-point review score drop, closing this regional gap is likely to directly raise satisfaction in these states.

3. Open a direct seller/quality review for Office Furniture

What: Audit packaging, damage-in-transit rates, and product-description accuracy for the Office Furniture category specifically (3.52 avg review vs. 4.16 marketplace average).

Supporting Analysis: Product Analytics.

Expected Benefit: This is a well-defined, bounded problem category rather than a marketplace-wide issue, making it a high-confidence quick win.

4. Build a dedicated account-management track for top-decile sellers

What: Formalize support (dedicated contact, priority issue resolution, early access to promotions) for the top 10% of sellers, who generate 67.1% of revenue.

Supporting Analysis: Seller Analytics.

Expected Benefit: Retention of even a few of these sellers protects disproportionate revenue relative to the cost of the program.

5. Reframe retention strategy around acquisition and first-purchase experience, not loyalty programs

What: Given a 97% one-time-purchase rate and near-zero month-over-month cohort retention, invest primarily in new-customer acquisition channels and first-order experience quality (packaging, on-time delivery, easy first-purchase checkout) rather than a traditional loyalty/rewards program.

Supporting Analysis: Cohort/Retention Analysis and Customer Analytics.

Expected Benefit: Avoids over-investing in retention mechanics that don't fit this marketplace's actual purchase pattern.

6. Set more conservative estimated-delivery dates in slow-performing regions

What: For states with historically long delivery times, consider adjusting the estimated delivery date shown to customers at checkout to better match real fulfillment times.

Supporting Analysis: Delivery & Logistics Analysis.

Expected Benefit: Where physical delivery speed cannot be quickly fixed, managing customer expectations accurately is a lower-cost lever that should reduce the 'late relative to promise' rate even before logistics improve.

7. Investigate the 1-star review segment specifically, not just the average score

What: Set up a recurring process (monthly, e.g.) to sample and read 1-star reviews (9.76% of all reviews) for recurring themes beyond delivery lateness.

Supporting Analysis: Customer Review Analysis.

Expected Benefit: A 4.16 average score can hide a real, addressable dissatisfied minority; treating them as a distinct group is more actionable than tracking the average alone.

Project Limitations

- **Historical time period:** the dataset covers September 2016 - October 2018 only. Findings may not reflect current marketplace dynamics, seasonality, or competitive conditions.
- **No profit or cost data:** the dataset contains price and freight (revenue) figures but no product cost or margin data. All 'revenue' figures in this report are exactly that — revenue, not profit — and should never be read as profitability.
- **Low repeat-purchase rate limits traditional retention analysis:** with only ~3% of customers purchasing more than once, month-over-month cohort retention percentages are small and can be noisy, especially for smaller/older cohorts. RFM segmentation was used as the primary customer-value framework for this reason.
- **Association, not causation:** the strong relationship between late delivery and low review scores is an association observed in the data, not a proven causal effect. Other unmeasured factors (product quality, communication, packaging) may also differ between on-time and late orders.
- **Customer identity limitations:** customer_unique_id is Olist's best available identifier for a real person, but if a customer used different contact details across orders, some repeat customers could be undercounted.
- **Marketplace-specific scope:** Olist is a multi-seller marketplace, not a single-brand retailer. Findings (e.g. seller concentration, category satisfaction) reflect marketplace-wide dynamics and may not generalize to a single-seller e-commerce business.

Conclusion

This analysis of the Olist e-commerce marketplace surfaced a consistent, evidence-backed picture: revenue and seller activity are concentrated (in São Paulo, in a handful of top categories, and among a small group of top sellers); customer growth is driven almost entirely by new-customer acquisition rather than repeat purchasing; and delivery reliability is the single clearest, most actionable lever available for improving customer satisfaction. Every recommendation in this report is tied directly to a specific, calculated finding from the underlying data — no assumptions, projections, or invented figures were used. Together, these findings and recommendations provide a realistic, business-focused foundation for a marketplace looking to grow revenue, improve satisfaction, and operate more efficiently.